A diagnostic framework for evaluating domain-informed graphs in urban flood identification

Abstract

Spatial relationships derived from domain knowledge, such as terrain flow directions, are increasingly incorporated into graph-based models for urban flood identification. However, improvements in aggregate predictive performance alone cannot establish whether these relationships provide information beyond each location's own attributes, how their gains depend on the way models use the graph, or where those gains occur. We propose a diagnostic framework encompassing four dimensions: reference comparison, assessment of modelling conditions, structural localization, and evaluation reassessment. Centred on predictive increments against matched references, the framework jointly examines their dependence on input information, predictor choice, upstream support, and evaluation settings. Applications to terrain-derived D8 graphs and flood records in Beijing and Zhengzhou showed that, under fixed propagation, ordinary spatial adjacency recovered most of the D8 gain over own attributes in Beijing. In Zhengzhou, a near-zero aggregate change concealed offsetting positive and negative contributions across upstream-support groups. Judgments of the incremental value of the same graph also varied with predictor choice and evaluation. The framework extends aggregate performance comparisons to multidimensional diagnosis, providing a systematic basis for evaluating and interpreting the predictive value of domain-informed graphs in urban flood-record identification.

1 Introduction

Where continuous water-level observations and calibrated drainage models are unavailable, urban flood identification commonly ranks locations using environmental attributes and historical flood records. Graph models connect related locations, allowing predictions to combine each node's own attributes with information from its neighbours. These connections can represent spatial relationships among urban units (Wang et al., 2023) or encode hydrological knowledge through river-network topology and catchment connectivity (Wang et al., 2025; Liu et al., 2026). On gridded terrain, the eight-direction flow-routing rule (D8) links each cell to its steepest downslope neighbour, representing terrain flow directions as directed upstream-to-downstream relationships (O'Callaghan & Mark, 1984). These applications raise a central question: how much predictive value do domain-informed connections add beyond own attributes and ordinary neighbourhood information?

Assessing this value requires quantitative comparisons against clearly specified data-driven references (Nearing et al., 2021). Wang et al. (2023), for example, compared the predictive performance of graph attention, artificial neural and convolutional neural networks. Such comparisons change both the use of neighbourhood information and the predictor architecture, making it difficult to attribute performance differences to a particular graph representation. An advantage over a model using only own attributes therefore does not establish an advantage over ordinary spatial adjacency; that distinction requires a corresponding reference comparison.

Graph increments may also depend on the information already available to each node and how the predictor uses its connections. Liu et al. (2026) distinguished directly supplied static attributes from information propagated through the graph and examined the effects of propagation range. Luan et al. (2023) analysed graph aggregation in terms of node distinguishability. Terrain information already encoded in node attributes may alter the increment available from the graph, while differences in how predictors use neighbourhood information can produce different comparative outcomes for the same connections. Input information and predictor choice are therefore conditions that need to be specified when interpreting a graph increment.

The spatial distribution of performance changes and the way they are evaluated also matter. Li et al. (2027) found that graph surrogates for urban drainage systems could exhibit substantial errors at local hydraulic boundaries despite high overall accuracy. Vemula et al. (2026) showed that model comparisons could differ between evaluations of pointwise prediction and spatial coherence. For flood-record identification, evaluation is further shaped by the observation process. Available records do not cover all inundated locations, so cells without records are treated as unlabelled background that may include unrecorded floods. Changing the composition or weight of this background can alter evaluation metrics and model rankings even when the predicted scores remain unchanged. Aggregate graph gains thus need to be interpreted alongside their local contributions and the conditions of evaluation.

Hydrological model diagnosis has examined structural differences through component comparisons under common conditions (Clark et al., 2008) and assessed the contributions of information sources through tiered comparisons (Ajao & Lim, 2026). Extending this approach to domain-informed graphs requires connecting reference choices, modelling conditions, structural contributions and evaluation to understand how they jointly shape judgments of graph value. We therefore propose a four-dimensional diagnostic framework comprising reference comparison, assessment of modelling conditions, structural localization, and evaluation reassessment. Centred on predictive increments against matched references, the framework systematically assesses the predictive value of domain-informed graphs. We test its diagnostic capabilities using synthetic data with known relationships and apply it to terrain-derived D8 graphs and flood records from the 2021 Zhengzhou and 2023 Beijing events.

This study makes two main contributions:

1. **A four-dimensional diagnostic framework.** The framework connects the reference against which a graph is assessed, the conditions under which its gains arise, the distribution of contributions across structural groups, and the dependence of comparative judgments on evaluation.
2. **Empirical evidence on graph increments obscured by aggregate comparisons.** Under fixed propagation, ordinary spatial adjacency recovered most of the D8 gain over own attributes in Beijing, while the near-zero aggregate change in Zhengzhou concealed offsetting group contributions. Graph increments also depended on predictor choice, and changes in evaluation could alter the ranking of models. These findings provide an empirical basis for interpreting the predictive value of domain-informed graphs.

2 Study areas and data

We study two city-event cases using flood-record windows from 19 July to 4 August 2021 in Zhengzhou and from 30 July to 11 August 2023 in Beijing (Figure 1). Each study area comprises the built-up land of the selected administrative unit and a surrounding 2 km buffer, delineated from ESA WorldCover and divided into 200 m cells. After excluding cells dominated by water or lacking adequate elevation data, the study areas contain 25,155 cells in Zhengzhou and 61,443 in Beijing.

Flood records come from CrowdFlood v1, which compiles flood inundation locations reported on Sina Weibo (Chen et al., 2026). We retained records in which every place name had a unique geocoding match and merged records falling in the same cell, which gave 791 recorded flood cells in Zhengzhou and 296 in Beijing. All other cells form an unlabelled background that can include unrecorded floods, consistent with the positive–unlabelled setting (Bekker & Davis, 2020). The task is retrospective ranking of recorded flood cells within each city-event case.

Each cell carries 18 environmental variables from the sources in Table 1. Terrain variables were derived from the Copernicus digital surface model (DSM) after aggregation to 200 m and depression filling (Barnes et al.,

2014), and the same filled surface defines the D8 graph (Section 3.2). The other variables were aggregated or assigned to the same cells, and no information from the flood reports enters the predictors. Supplementary Text S1 gives the processing details.

Table 1. Data sources.

Data	Source	Native resolution
Flood records	CrowdFlood v1 (Chen et al., 2026)	Point locations
Terrain	Copernicus DEM GLO-30 (European Space Agency & Airbus, 2022)	30 m
Rainfall	GPM Integrated Multi-satellitE Retrievals (IMERG), Final V07B (Huffman et al., 2023)	0.1°, 30 min
Land cover	ESA WorldCover 2021 v200 (Zanaga et al., 2022)	10 m
Roads and waterways	OpenStreetMap (OSM) history via ohsome (Raifer et al., 2019)	Vector
Population	WorldPop Global2 R2024B, unconstrained (WorldPop, 2025)	100 m
Administrative units	geoBoundaries ADM2 (Runfola et al., 2020)	Vector

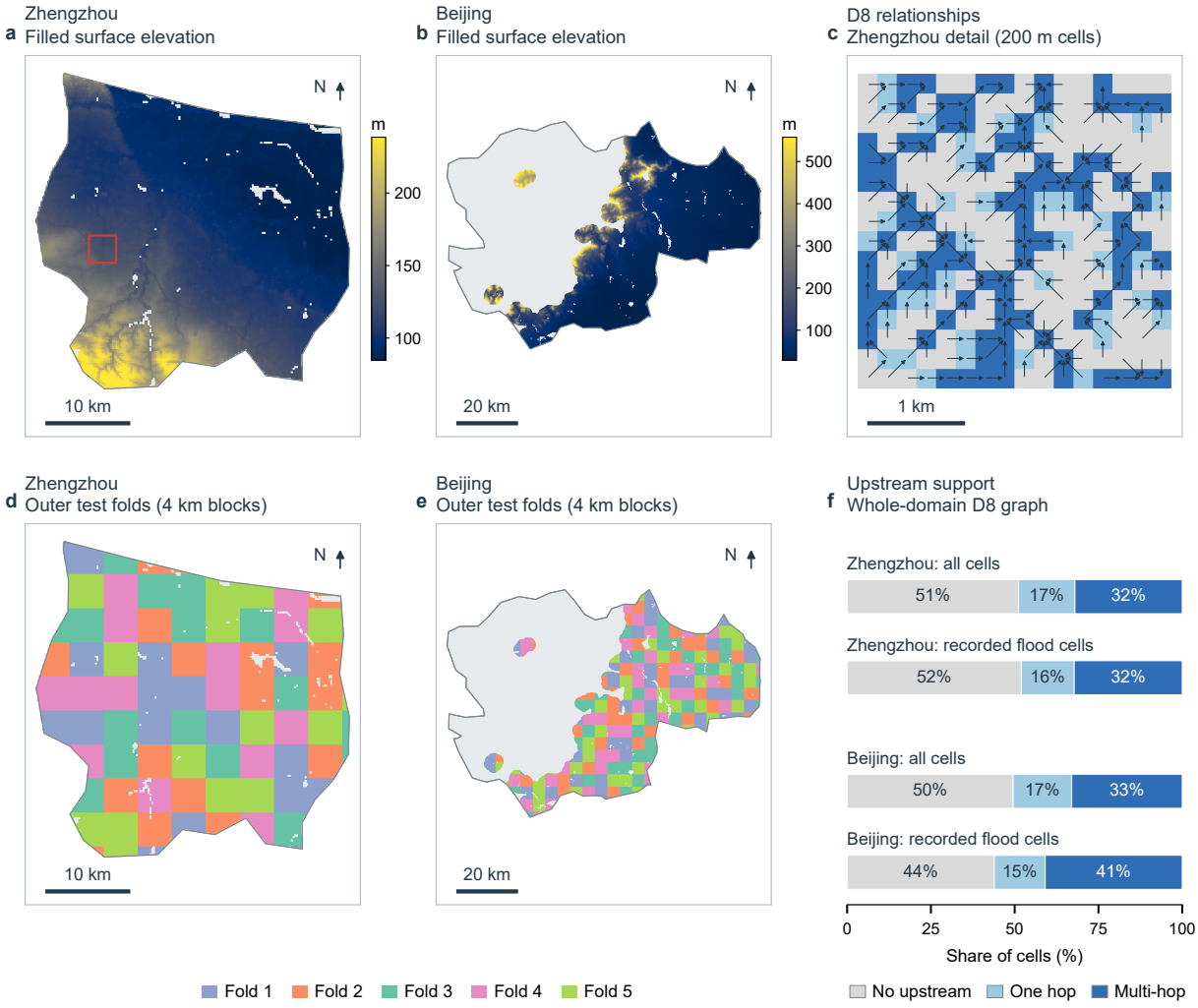


Figure 1. Study areas and spatial design. (a, b) Depression-filled surface elevation on the 200 m grids; pale grey marks administrative area outside the study extent. (c) D8 flow directions in the window marked in (a), with cells coloured by upstream support. (d, e) Assignment of 4 km blocks to the five outer test folds. (f) Shares of upstream-support groups among all cells and among recorded flood cells, computed on the whole-domain graph.

3 Methods

3.1 Diagnostic framework

The framework evaluates the predictive contribution of a domain-informed graph through matched comparisons, then examines how that contribution depends on information availability, predictor choice, structural support and evaluation (Figure 2). Comparisons with own attributes and ordinary adjacency establish the increment to be examined. Attribute removal and predictor comparisons assess the conditions under which it arises. Structural decomposition locates gains and losses within the overall change, while reweighting the same predictions tests the dependence on evaluation. Together, these analyses connect the reference, conditions, location and evaluation of a graph increment. We apply the framework to D8 relationships and use controlled synthetic data to test the recovery of useful relationships and support losses.

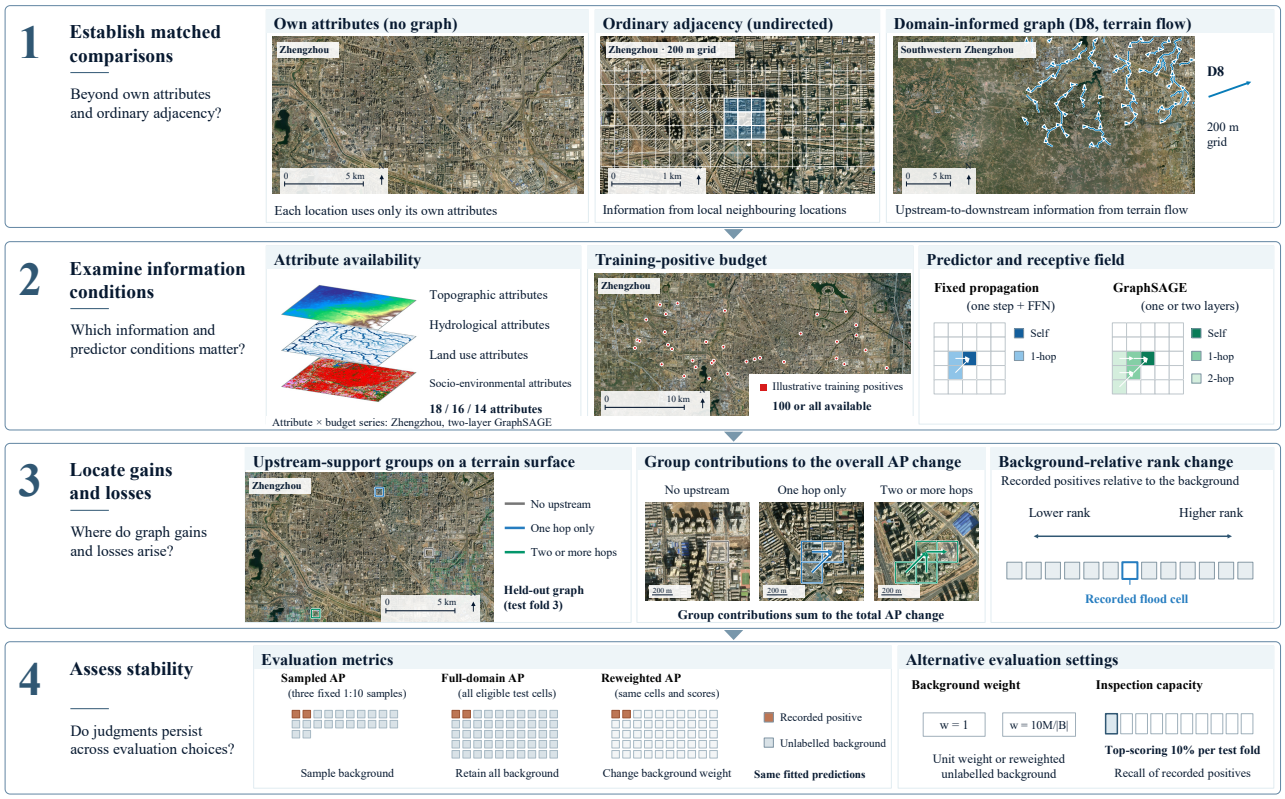


Figure 2. Four diagnostic dimensions for evaluating domain-informed graphs, illustrated on the Zhengzhou 200 m grid. (1) Compare D8 with own attributes and ordinary adjacency under matched fitting. (2) Examine attribute availability, training-positive budget, predictor and propagation depth; the 18/16/14-attribute and 100/all-positive comparisons use two-layer GraphSAGE in Zhengzhou. (3) Allocate the full-domain AP difference to no-upstream, one-hop-only and multi-hop support groups in the held-out graph, and examine recorded-positive ranks relative to background. Group AP contributions are additive; rank changes are not. (4) Reassess the same predictions using sampled, full-domain and reweighted AP, and recorded-positive recall within the top 10% of cells in each test fold. M and $|B|$ denote recorded-positive and background counts; w is the background weight. Geographic panels provide spatial context, with support examples from test fold 3; training-positive locations and evaluation diagrams are illustrative. AP, average precision; FFN, feed-forward network.

3.2 Matched comparisons of graph information and its use

We represent the study areas as graphs whose nodes are the 200 m cells described in Section 2. The own-attribute reference has no external edges, and ordinary adjacency connects each cell to its eight neighbours in both directions. D8 connects a cell to its steepest downslope neighbour on the depression-filled surface (O'Callaghan & Mark, 1984); drainable flats are routed towards outlets. These directed edges carry information from upstream to downstream cells. Comparing D8 with own attributes measures its predictive increment beyond local inputs;

comparing it with ordinary adjacency evaluates two ways of selecting and weighting neighbouring attributes. These are comparisons of complete graph representations under matched fitting protocols.

To make these comparisons within the same predictor family, fixed propagation supplies a feed-forward network with both the cell's own attributes and a neighbourhood average. For standardised attributes X and incoming neighbours $\mathcal{N}_i^-$,

$$(HX)_i = \frac{X_i + \sum_{j \in \mathcal{N}_i^-} X_j}{1 + |\mathcal{N}_i^-|}, \qquad Z = [X, HX].$$

For the own-attribute reference, $H = I$ and the input is $[X, X]$. Thus, the three relations use the same 36-dimensional input and candidate network architectures in the main comparisons. We also compare the three relations using two-layer GraphSAGE, which learns transformations of each cell's representation and its incoming-neighbour mean (Hamilton et al., 2017). On the D8 graph, comparing fixed propagation with GraphSAGE assesses dependence on the predictor. A one-layer GraphSAGE variant matches the direct-neighbour reach of one-step fixed propagation; the two-layer variant extends the comparison to two-hop propagation.

Information already encoded in the attributes may reduce the increment available from the graph. We examine this possibility in the Zhengzhou two-layer GraphSAGE experiments by removing log flow accumulation and the topographic wetness index, then also relative elevation and local relief. The resulting attribute sets contain 18, 16 and 14 variables. Each is evaluated with 100 revealed training positives and with all available training positives. We compare the D8 increment over own attributes before and after removal, then compare that change between label budgets. These contrasts measure the response to the specified attribute removals at each label budget. In this separate ablation series, the own-attribute and adjacency models inherit the configuration and training duration selected for the D8 model within each attribute set and budget. The budget interaction and direct removal effects are defined in Supplementary Text S2 and reported with their foldwise values in the supporting results.

3.3 Structural decomposition and evaluation dependence

Average precision (AP) measures how well recorded flood cells rank above the unlabelled background. Let P contain the M recorded flood cells, B the background cells, and s_i the predicted score of cell i. With background weight w relative to unit weight for each recorded flood cell,

$$p_i(w) = \frac{\#\{j \in P : s_j \geq s_i\}}{\#\{j \in P : s_j \geq s_i\} + w\,\#\{j \in B : s_j \geq s_i\}}, \qquad \mathrm{AP}(w) = \frac{1}{M} \sum_{i \in P} p_i(w).$$

The threshold includes the complete block of tied scores. Primary comparisons use all recorded flood cells against background samples ten times their number, with $w = 1$, averaged over three fixed samples. This fixed evaluation mixture matches the inner-selection criterion and gives the same nominal recorded-positive weight across folds and cities. Precision–recall measures depend on this mixture (Saito & Rehmsmeier, 2015); full-domain and reweighted evaluations therefore accompany the sampled comparisons.

To reveal changes concealed by the total AP difference, we group recorded flood cells by their upstream support in the held-out graph: no incoming edge, one-hop support without a two-hop path, or support through a path of at least two hops. These groups describe graph structure independently of predictor depth. For predictions a and b, the contribution of group g is

$$C_g = \frac{1}{M} \sum_{i \in P_g} [p_i^a(1) - p_i^b(1)], \qquad \sum_g C_g = \mathrm{AP}_a(1) - \mathrm{AP}_b(1).$$

We calculate these terms on the full test domain, retaining all candidates in the ranking. The contributions therefore account exactly for the overall change, including offsets between groups. Each positive cell's precision depends on the complete candidate ranking, and the compared models are fitted separately. The decomposition assigns their ranking difference to structural groups; it does not isolate the effect of the edges incident to those cells. Dividing C_g by the group's share of positives gives its mean precision change. To distinguish this precision contribution from movement relative to background cells, we also calculate

$$r_i = \frac{\#\{j \in B : s_j < s_i\} + 0.5\,\#\{j \in B : s_j = s_i\}}{|B|}.$$

Group means of $r_i^a - r_i^b$ show whether recorded flood cells move ahead of the background. These rank changes and the AP contributions describe complementary aspects of the same fitted predictions.

To assess variation between refits, we also compare archived own-attribute models that differ only in neural random seed. These seed-only comparisons retain the partition, revelation draw, selected setting, support groups and complete candidate ranking. Fixed-pair and randomized references, together with leave-one-refit-slot-out graph contrasts, assess the sensitivity of the structural contributions; Supplementary Text S2 gives their averaging and reference-distribution definitions.

We then examine whether comparative judgments depend on the evaluation mixture. Full-domain AP includes every eligible test cell with $w = 1$. Reweighted full-domain AP retains those cells and scores but sets $w = 10M/|B|$, giving the background ten times the total weight of recorded flood cells. Comparing these two measures isolates the effect of weighting; comparison with sampled AP additionally reflects the choice of background candidates. Recorded-positive recall is evaluated among the top-scoring 10% of cells separately within each test fold, using 10% as an illustrative inspection capacity. This allocates a quota to each held-out region rather than distributing one citywide budget through a pooled ranking. All evaluations use the selected models without further fitting or selection.

3.4 Validation with known relationships

Controlled synthetic graphs allow diagnostic comparisons with known relationships (Palowitch et al., 2022). Here, we test whether the framework distinguishes useful from unrelated relationships and locates losses caused by removed support. We generate ten independent grids with three standard-normal attributes per cell and horizontal directed chains of four cells. Labels follow

$P(y_i = 1) = \sigma(-2 + 0.5x_{i1} + 2\bar{x}_{\uparrow i1})$,

where $\bar{x}_{\uparrow i1}$ is the mean first attribute of incoming neighbours, set to zero when none are present, and σ is the logistic function.

The useful chains are compared with unrelated vertical adjacency, degree-preserving rewiring and boundary truncation. An additional condition retains the chains but generates labels independently with probability 0.2. These conditions respectively examine relationship relevance, connectivity without the original relationships, loss of support and behaviour without predictive signal. Logistic models use either own attributes or own attributes plus the incoming-neighbour mean, with regularisation selected on spatially separated validation data. Recovery is assessed through paired AP increments and by comparing truncation losses between cells that lose

support and those that retain it. Ranges give the observed minimum and maximum across the ten grids. Grid construction and fitting details are provided in Supplementary Text S3.

3.5 Spatial validation and estimation

Both cities use nested spatial validation with 4 km blocks, five outer folds and three inner folds. A 1 km buffer separates training and evaluation cells, and graph edges crossing partitions or touching buffer cells are removed. Preprocessing is fitted on the training sample; hyperparameters and training duration are selected using inner-fold sampled AP. The main model variants are selected independently from the same two candidate settings, and the median best inner-fold epoch determines each outer refit's duration. These rules evaluate ranking in spatially withheld areas of each event under the stated model-selection budget. The supplied computational record identifies the configurations, seeds, selection outputs and analysis entry points.

The main comparisons use a common limited-supervision budget of 100 revealed training positives in each city, under five revelation draws, with 2,000 unlabelled training cells per draw and three neural refit seeds. Revealed positives and unlabelled cells receive equal total weight in the training loss. Within each comparison, models are paired by outer spatial fold and, where applicable, revelation draw, background sample and refit-seed slot. Network settings, preprocessing and training details are retained in Supplementary Text S2.

For each comparison, we average repeated evaluations within each outer fold and then give the five folds equal weight. Descriptive ranges are the 2.5th and 97.5th percentiles of 10,000 paired resamples of the outer folds, accompanied by the number of folds with positive increments. Common resampling draws preserve pairing, including both budgets in the attribute interaction. These ranges describe variation across the five folds of the fixed spatial partition, conditional on the fitted predictions. They are not confidence intervals for transfer to other partitions or events. Equal-fold averages give each held-out region the same weight and differ from metrics computed after pooling all test-cell scores. AP and recall differences are reported in percentage points; background-rank changes are reported as 100 times the rank difference.

4 Results

Unless specified otherwise, differences are in sampled AP percentage points (pp), and brackets give descriptive ranges from paired outer-fold resampling. Table 2 reports absolute main-model performance and Table 3 the attribute-removal comparisons. Supplementary Table S3 gives fold denominators; the accompanying CSVs retain foldwise metrics and paired differences.

4.1 D8 gains depend on the spatial reference

Fixed D8 propagation improved the ranking of recorded flood cells in Beijing relative to own attributes, with a mean AP increment of 2.54 pp [1.13, 4.40] and positive increments in all five spatial folds (Figure 3). Mean sampled AP was 31.38% for D8, 28.84% for own attributes and 30.73% for adjacency. The increment over ordinary adjacency was 0.65 pp [−0.28, 1.44], with positive values in four folds. Much of the gain over own attributes was therefore also available from ordinary neighbourhood information. The two references distinguish a positive spatial-context increment in all five folds from the smaller, fold-dependent difference between the D8 and adjacency representations.

The same comparison in Zhengzhou produced a different pattern, with sampled AP of 28.37% for D8, 28.41% for own attributes and 28.94% for adjacency. The D8 increment over own attributes was −0.03 pp [−0.51, 0.42],

while its increment over adjacency was −0.57 pp [−1.08, 0.14]. Thus, the Beijing result did not extend to both city-event cases under the same predictor and evaluation.

Table 2. Absolute performance of the main models. AP and recall are percentages, averaged equally over the five spatial folds after within-fold averaging. Fixed propagation uses one step; Own denotes the own-attribute reference. Reweighted AP uses the full test domain with a 1:10 recorded-positive-to-background total weight. Recall uses the top-scoring 10% separately within each test fold.

City	Predictor	Relation	Sampled AP	Full-domain AP	Reweighted AP	Recall
Beijing	Fixed propagation	Own	28.837	2.253	28.776	41.649
Beijing	Fixed propagation	Adjacency	30.725	2.533	30.284	42.576
Beijing	Fixed propagation	D8	31.379	2.606	30.834	44.479
Beijing	GraphSAGE, 2 layers	Own	29.590	2.375	29.644	42.400
Beijing	GraphSAGE, 2 layers	Adjacency	28.560	2.429	28.360	39.292
Beijing	GraphSAGE, 2 layers	D8	28.103	2.584	28.112	38.123
Beijing	GraphSAGE, 1 layer	D8	29.046	2.555	29.054	40.593
Zhengzhou	Fixed propagation	Own	28.407	11.916	28.456	38.580
Zhengzhou	Fixed propagation	Adjacency	28.941	12.251	29.039	39.595
Zhengzhou	Fixed propagation	D8	28.373	11.919	28.457	38.639
Zhengzhou	GraphSAGE, 2 layers	Own	28.622	12.170	28.775	38.525
Zhengzhou	GraphSAGE, 2 layers	Adjacency	28.690	12.171	28.835	38.918
Zhengzhou	GraphSAGE, 2 layers	D8	25.995	10.748	26.198	35.180
Zhengzhou	GraphSAGE, 1 layer	D8	27.045	11.207	27.107	36.813

4.2 Predictor dependence and mixed responses to attribute removal

The positive Beijing increment depended on how the graph entered the predictor. With two-layer GraphSAGE, the mean D8 increment over own attributes was −1.49 pp [−3.36, 0.63], with positive values in two folds (28.10% versus 29.59% AP). In Zhengzhou, the corresponding increment was −2.63 pp [−4.59, −0.64] (Figure 3). Under the tested architectures and selection budget, the positive result for fixed D8 propagation in Beijing did not transfer to two-layer GraphSAGE.

Predictor differences were already present when neighbourhood reach was matched. When both predictors used one-hop D8 neighbourhoods, fixed propagation exceeded one-layer GraphSAGE by 2.33 pp in Beijing and 1.33 pp in Zhengzhou. Both descriptive ranges were above zero. The difference persisted when GraphSAGE used two layers, reaching 3.28 and 2.38 pp, respectively.

Terrain-attribute removal addressed whether withholding terrain summaries increased the D8 increment over own attributes. This separate Zhengzhou GraphSAGE series used D8-selected settings for its own-attribute references. With 100 revealed positives, removing flow accumulation and wetness changed the increment by −0.06 pp relative to the full attribute set; also removing relative elevation and local relief changed it by −0.15 pp. With all training positives revealed, the corresponding changes were +0.03 and +0.19 pp. The mean responses therefore changed direction between budgets, while both paired budget interactions varied in sign across spatial folds and had descriptive ranges spanning zero. For these Zhengzhou GraphSAGE fits, attribute removals and shared selection settings, no consistent increase in the D8 increment was observed. Table 3 reports the absolute AP and graph increments; direct removal effects are supplied fold by fold in the accompanying CSVs.

4.3 A near-zero overall change conceals opposing structural contributions

The near-zero Zhengzhou increment combined distinct changes across upstream-support groups. Under full-domain evaluation, fixed D8 propagation changed AP by +0.003 pp relative to own attributes. Recorded flood cells without incoming support contributed +0.181 pp, those with one-hop support contributed +0.004 pp, and

Table 3. Attribute removal and the D8 increment in the separate Zhengzhou two-layer GraphSAGE series. Own and D8 columns give sampled AP (%); increments and their changes are percentage points. Brackets retain the descriptive paired-fold resampling ranges. The 16-attribute set removes flow accumulation and wetness; the 14-attribute set also removes relative elevation and local relief. All denotes all available training positives. Own-attribute fits inherit the D8-selected setting and duration within each row.

Training positives	Attributes	Own AP	D8 AP	D8 minus own	Change from 18
100	18	27.542	25.995	-1.547 [-2.990, -0.230]	Reference
100	16	28.906	27.299	-1.607 [-2.982, -0.375]	-0.060
100	14	29.007	27.312	-1.695 [-2.811, -0.563]	-0.148
All	18	30.395	29.061	-1.333 [-2.293, +0.096]	Reference
All	16	30.604	29.304	-1.300 [-3.637, +1.378]	+0.034
All	14	30.641	29.503	-1.138 [-2.497, +0.447]	+0.195

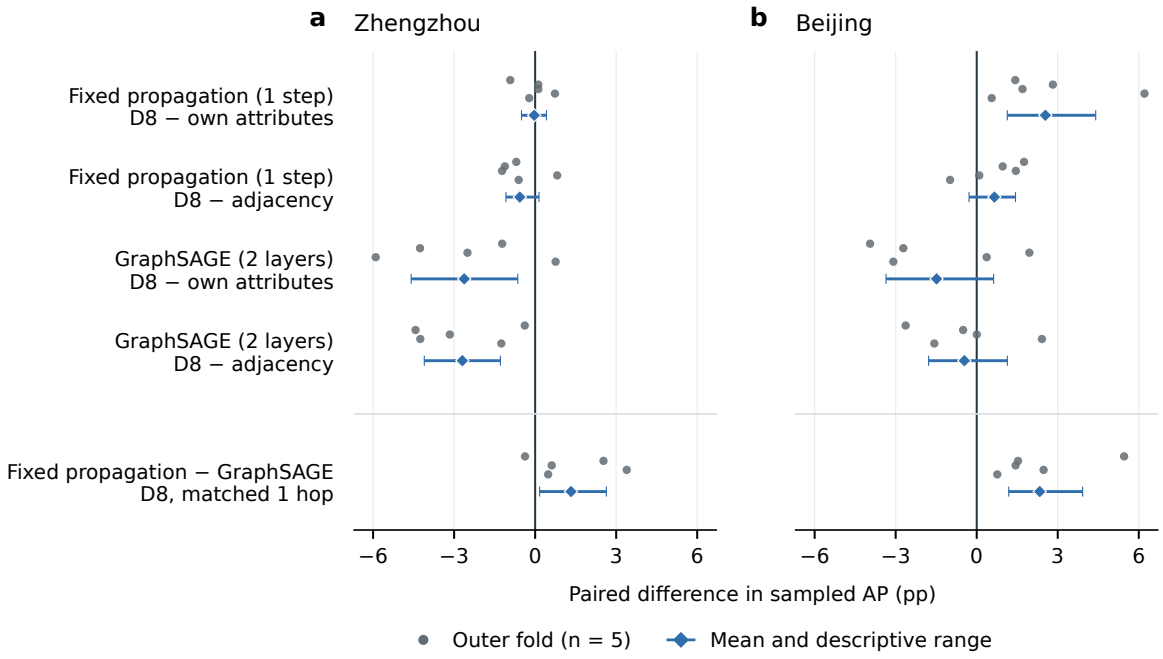


Figure 3. Matched comparisons of D8 relationships and predictor use. Panels show Zhengzhou (a) and Beijing (b). The first four comparisons evaluate D8 against own attributes or ordinary spatial adjacency within the same predictor family. The final comparison holds D8 fixed and compares one-step fixed propagation with one-layer GraphSAGE; the other GraphSAGE comparisons use two layers. Grey circles show five outer-fold paired differences after averaging neural refit seeds and revelation draws within each fold. Sampled AP averages three fixed 1:10 background samples. Blue diamonds indicate the equally weighted fold mean; whiskers show the descriptive 2.5th and 97.5th percentiles from 10,000 paired fold resamples. Vertical offsets separate fold points from summaries and carry no numerical meaning. Differences are expressed in percentage points.

those with multi-hop support contributed −0.182 pp (Figure 4a). The near-zero total arose from offsetting precision contributions assigned to these groups within the complete ranking.

The multi-hop contribution was negative in every spatial fold, with a descriptive range of [−0.355, −0.028] pp (Figure 4b). Its mean background-rank change was also negative, at −0.93 pp [−1.51, −0.25], showing that these recorded flood cells moved behind a larger share of background cells (Figure 4c). The paired mean precision change favoured the one-hop group over the multi-hop group by 0.64 pp [0.16, 1.27], with the same direction in all five folds. Together, the decomposition and rank diagnostic locate the adverse change within a structurally defined group that the aggregate score conceals.

Beijing showed positive mean AP contributions from all three support groups, summing to a full-domain increment of 0.353 pp. The multi-hop group contributed +0.189 pp [0.063, 0.325], with positive contributions in all five folds. The same structural category therefore accompanied gains in Beijing and losses in Zhengzhou.

Upstream support identifies where the increment arises, while the two cases show that more extensive support does not imply a uniform predictive benefit.

Seed-only comparisons assess how this allocation varies between refits (Table 4). The Zhengzhou multi-hop contribution lies outside the fixed-pair envelope of [-0.081, +0.081] pp but within the randomized seed-reference range of [-0.221, +0.223] pp. Omitting each of the three refit slots in turn leaves a negative city mean, from -0.212 to -0.163 pp, with negative contributions in four or five folds. The negative mean therefore persists under these omissions, while the randomized reference places its magnitude within the variation observed between own-model refits. These comparisons describe refit sensitivity; the reference uses one seed pair per unit, whereas the graph contrast averages three refits per model.

Table 4. Structural contributions and seed-only sensitivity references, in full-domain AP percentage points. The graph contrast averages three refits of each model. Fixed-pair ranges span all six signed own-model seed-pair means; randomized ranges are the 2.5th-97.5th percentiles of 10,000 draws selecting one ordered pair independently per fold-revelation unit. Both references average five revelations and then five folds equally. These descriptive ranges are not confidence intervals or a variance-matched test of the graph contrast.

City	Support	D8 minus own	Fixed-pair range	Randomized range
Beijing	No upstream	+0.148	[-0.052, +0.052]	[-0.057, +0.056]
Beijing	One hop	+0.017	[-0.012, +0.012]	[-0.019, +0.019]
Beijing	Multi-hop	+0.189	[-0.135, +0.135]	[-0.194, +0.195]
Zhengzhou	No upstream	+0.181	[-0.101, +0.101]	[-0.226, +0.231]
Zhengzhou	One hop	+0.004	[-0.054, +0.054]	[-0.085, +0.084]
Zhengzhou	Multi-hop	-0.182	[-0.081, +0.081]	[-0.221, +0.223]

These groups use the held-out graphs. Of the whole-domain multi-hop recorded positives, 233 of 255 in Zhengzhou and 112 of 121 in Beijing retained that membership after partitioning; none entered the group. The reported multi-hop contributions already concern these retained members. Tables S3–S4 document the transitions, group sizes and removed incoming edges.

4.4 Evaluation changes which comparisons remain consistent

Re-evaluating the fitted predictions distinguished persistent graph gains from comparisons sensitive to background weighting. In Beijing, fixed D8 propagation remained above the own-attribute reference in every fold under sampled, full-domain and reweighted full-domain AP. Full-domain AP was 2.606% for D8 and 2.253% for own attributes. Its mean increment was 0.35 pp [0.23, 0.52] on the full domain and 2.06 pp [1.09, 3.63] after reweighting. The size of the increment changed with evaluation, while its direction persisted across folds.

The comparison between fixed propagation and two-layer GraphSAGE was more sensitive. Their Beijing difference was 3.28 pp under sampled AP, 0.02 pp [−0.33, 0.42] under full-domain AP (2.606% versus 2.584%), and 2.72 pp [1.54, 3.99] under reweighted full-domain AP (30.834% versus 28.112%). Fixed propagation ranked higher in two folds under full-domain AP and in all five after reweighting. Because the latter two evaluations use identical cells and scores, this change in foldwise ordering follows from the weight assigned to the background.

A fixed inspection capacity supplied a complementary judgment about the reference comparison. Within the top-scoring 10% of cells in each test fold, D8 propagation increased recorded-positive recall relative to adjacency by 1.90 pp [0.53, 3.69] in Beijing, from 42.58% to 44.48%. In Zhengzhou, recall changed from 39.59% to 38.64%, a difference of −0.96 pp [−1.33, −0.56], with a negative value in every fold. In Beijing, the recall and sampled-AP contrasts favoured D8 in the same four folds, but only the recall contrast had a descriptive range wholly above zero.

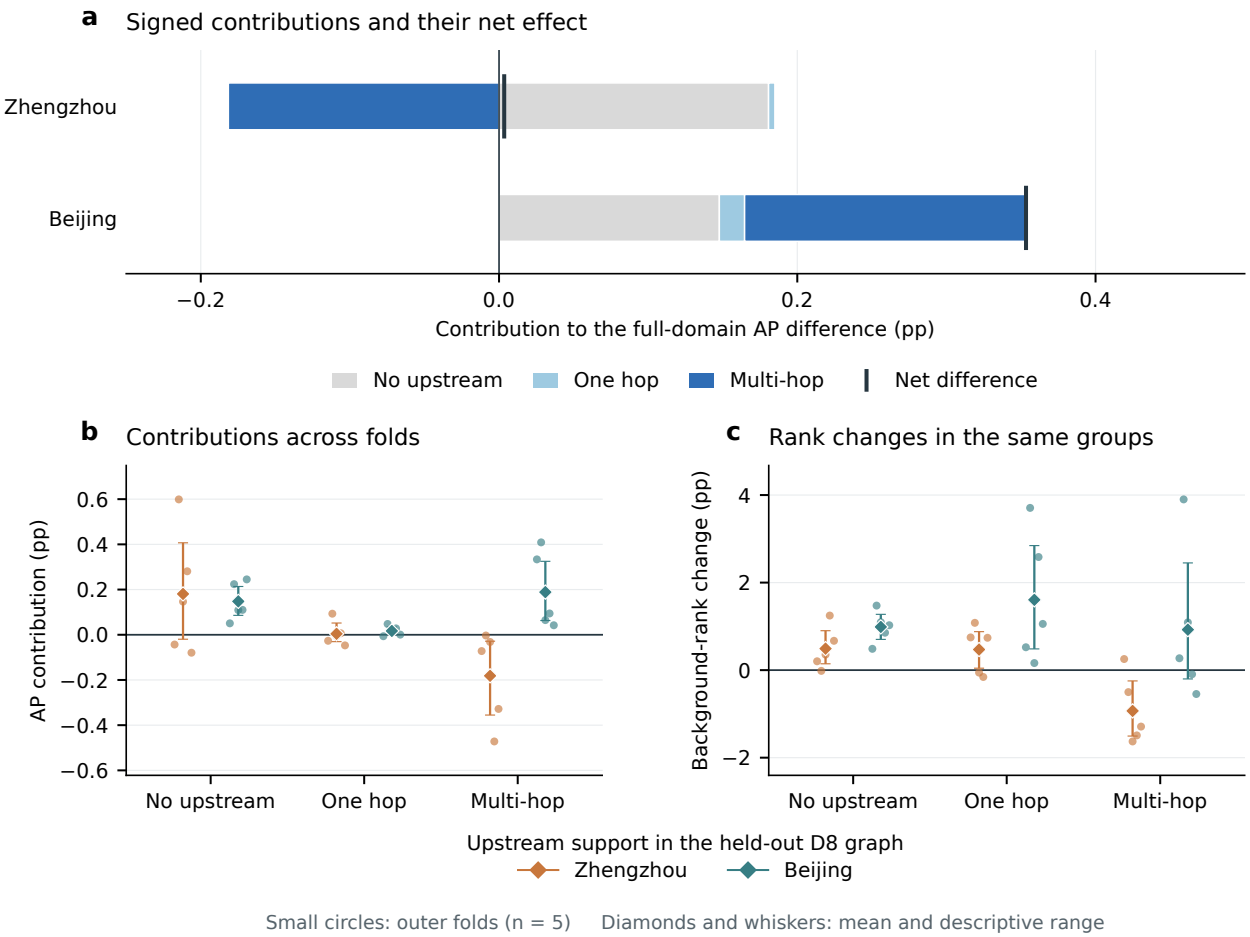


Figure 4. Structural decomposition and background-rank changes for one-step fixed D8 propagation relative to the own-attribute reference. All panels use full-domain evaluation and groups defined by incoming support in the held-out D8 graph: none, one hop without a two-hop path, and a path of at least two hops. Panel a stacks signed mean AP contributions on either side of zero; black ticks indicate the net paired difference. Contributions sum to the net difference within each fold and in the displayed means. Panel b shows AP contributions by group and city; panel c shows background-rank changes in the same groups. Small circles represent the five outer folds, diamonds their means, and whiskers the descriptive resampling ranges defined in Figure 3. The groups describe held-out graph structure independently of predictor depth. Precision and background ranks are computed against the complete test-candidate set; contributions allocate the paired model difference and are not effects of incident edges in isolation. AP contributions are in percentage points; background-rank changes are normalized differences multiplied by 100. Group mean rank changes are not additive.

4.5 Known relationships and support losses are recovered in synthetic data

The controlled data established that the reference and location diagnostics responded to the relationships built into the generator. Useful relationships increased AP over own attributes in all ten independent grids, with a mean gain of 37.52 pp and a replicate range of 33.73–40.64 pp. Mean gains for unrelated adjacency, degree-preserving rewiring and independent labels were −0.03, +0.12 and +0.32 pp, respectively, with both positive and negative values across replicates. Preserving node degrees through rewiring did not preserve the gain from the useful relationships.

Removing support reduced AP relative to the useful graph in every grid, by a mean of 4.77 pp and a replicate range of 2.44–6.57 pp. Cells whose support was removed had more adverse precision and background-rank changes than cells retaining support in all ten grids. These results demonstrate, within the specified synthetic setting, the two diagnostic capabilities used to examine D8: distinguishing useful relationships from unrelated edges and locating losses associated with the removal of support.

5 Discussion

Assessing the predictive contribution of a domain-informed graph requires distinguishing the gains from introducing neighbourhood information from the additional value of a particular connection pattern. In Beijing, ordinary adjacency recovered most of the improvement that fixed D8 propagation achieved over own attributes. Using own attributes as the sole reference would attribute this neighbourhood gain to D8. An ordinary-adjacency reference allows a further assessment of whether organizing information along potential terrain flow paths offers more predictive value than spatial proximity alone. This connects the quantitative assessment of domain knowledge (Nearing et al., 2021) to the choice of graph representation. Because D8 and ordinary adjacency differ in neighbour composition, directionality and aggregation weights, their contrast evaluates the two representations as a whole.

The value of these connections also depends on how their information enters prediction. The graph determines which locations can exchange information, while the predictor determines how that information is combined with each node's own attributes. The contrasting performance of fixed propagation and GraphSAGE shows that the same connections need not yield the same predictive gains. A gain under one predictor therefore provides a conditional basis for adopting the graph. In Zhengzhou, removing explicit terrain attributes did not consistently increase the graph increment under the tested feature-removal and model-selection protocol. Information overlap between terrain attributes and graph relationships thus remains a question to be examined under specific input and learning configurations. Graph structure, input attributes and predictor choice should consequently be evaluated together.

Even with these modelling conditions specified, the aggregate increment can obscure changes across locations. The near-zero change in Zhengzhou contained offsetting group contributions, showing that little overall improvement can coexist with differences in ranking performance. Decomposition by upstream support relates these differences to nodes' positions within the terrain-derived graph and identifies structural groups for further investigation. The decomposition locates differences between separately fitted models; the negative contribution of the multi-hop group remained within the range of the randomized seed-only reference. The evidence therefore identifies a group difference whose persistence requires further examination. The magnitude of the aggregate increment and the distribution of group contributions are distinct aspects of graph performance.

Evaluation determines how these changes enter the performance comparison. Reweighting the background can alter comparative judgments without changing predicted scores, showing that the apparent advantage of a graph also depends on the relative weight assigned to recorded positives and unlabelled background. AP over the full candidate set and recorded-positive recall within the top 10% of cells in each test fold assess overall ranking and retrieval under limited capacity, respectively, and may favour different choices. Evaluation reassessment should therefore follow an explicit intended use, rather than select whichever metric gives the most favourable result. For social-media flood records, these metrics assess the ranking of recorded locations; background reweighting does not correct spatial differences in reporting propensity.

Support for a particular graph representation therefore depends on what it adds beyond ordinary neighbourhood information, whether that increment is realised by the chosen predictor, and how the gains align with the structural groups and evaluation objectives of interest. The framework connects these judgments through corresponding comparisons, giving both positive gains and near-zero changes a more specific interpretation. The present empirical evidence concerns flood-record ranking in two city events, with 200 m D8 graphs representing potential terrain connectivity. Future studies can retain this comparative design while extending it to additional events and graph representations to examine whether the identified conditions for gains and differences among groups recur.

6 Conclusions

We developed a four-dimensional diagnostic framework that extends the evaluation of domain-informed graphs beyond aggregate performance comparisons to a systematic analysis of predictive gains. Applications in Beijing and Zhengzhou identified gains also attainable through ordinary spatial adjacency and revealed differences among structural groups that aggregate metrics obscured. Further analyses showed that the incremental value of the same graph depended on predictor choice, while changes in evaluation could alter model rankings. These findings provide a basis for interpreting the predictive value of domain-informed graphs under clearly specified modelling and evaluation conditions. Future work should examine the framework's applicability across additional flood events and graph types.

Data and code availability

The accompanying review package contains source code and frozen configurations with file-level checksums, dependency specifications, model-selection records, aggregate results and figure-generation scripts. The input–output guide links the comparisons to their analysis entry points. Original datasets should be obtained from the providers cited in Table 1 using the version identifiers documented in Supplementary Text S1. Individual social-media records and location-level derived training and prediction data are excluded from this package. Aggregate tables provide an independent route to check the reported contrasts, while reproducing model fitting requires the provider-acquired inputs and prepared spatial state described in the guide. A durable public archive and code licence have not yet been assigned to this review package.

References

Ajao, A., & Lim, Y. H. (2026). Diagnosing the Roles of Meteorological Forcing, Storage Memory, and Network Connectivity in Cold-Region Streamflow Prediction. *Water Resources Management, 40*(11), 513. https://doi.org/10.1007/s11269-026-04869-9

Barnes, R., Lehman, C., & Mulla, D. (2014). Priority-flood: An optimal depression-filling and watershed-labeling algorithm for digital elevation models. *Computers & Geosciences, 62*, 117-127. https://doi.org/10.1016/j.cageo.2013.04.024

Bekker, J., & Davis, J. (2020). Learning from positive and unlabeled data: A survey. *Machine Learning, 109*(4), 719-760. https://doi.org/10.1007/s10994-020-05877-5

Chen, Y., Liu, G., Yuan, Z., Lyu, Z., Hu, M., & Chen, X. (2026). *A dataset of flood inundation locations from social media in China* (v1) [Data set]. Zenodo. https://doi.org/10.5281/zenodo.19921295

Clark, M. P., Slater, A. G., Rupp, D. E., Woods, R. A., Vrugt, J. A., Gupta, H. V., Wagener, T., & Hay, L. E. (2008). Framework for Understanding Structural Errors (FUSE): A modular framework to diagnose differences between hydrological models. *Water Resources Research, 44*(12), W00B02. https://doi.org/10.1029/2007WR006735

European Space Agency & Airbus. (2022). *Copernicus DEM* [Data set]. European Space Agency. https://doi.org/10.5270/ESA-c5d3d65

Hamilton, W., Ying, Z., & Leskovec, J. (2017). Inductive Representation Learning on Large Graphs. *Advances in Neural Information Processing Systems, 30*. NeurIPS proceedings

Huffman, G. J., Stocker, E. F., Bolvin, D. T., Nelkin, E. J., & Tan, J. (2023). *GPM IMERG Final Precipitation L3 Half Hourly 0.1 degree x 0.1 degree V07* [Data set]. NASA Goddard Earth Sciences Data and Information Services Center. https://doi.org/10.5067/GPM/IMERG/3B-HH/07

Li, L., Li, L., Dong, Q., Bai, S., Ho, S.-H., Zhao, X., Di, D., Li, J., Wang, X., Xu, Y., Yang, T., & Ren, N. (2027). Where accurate graph neural network surrogates fail in urban drainage systems: Hidden local failures under overflow transitions. *Water Research, 308*(Part C), 127021. Published online 25 September 2026. https://doi.org/10.1016/j.watres.2026.127021

Liu, J., Zhao, G., Xiong, J., & Kinouchi, T. (2026). A Graph-Based Deep Learning Approach for Daily Flash Flood Susceptibility Modeling in China. *Water Resources Research, 62*(4), e2025WR041360. https://doi.org/10.1029/2025WR041360

Luan, S., Hua, C., Xu, M., Lu, Q., Zhu, J., Chang, X.-W., Fu, J., Leskovec, J., & Precup, D. (2023). When Do Graph Neural Networks Help with Node Classification? Investigating the Homophily Principle on Node Distinguishability. *Advances in Neural Information Processing Systems, 36*, 28748-28760. https://doi.org/10.52202/075280-1251

Nearing, G. S., Kratzert, F., Sampson, A. K., Pelissier, C. S., Klotz, D., Frame, J. M., Prieto, C., & Gupta, H. V. (2021). What Role Does Hydrological Science Play in the Age of Machine Learning? *Water Resources Research, 57*(3), e2020WR028091. https://doi.org/10.1029/2020WR028091

O'Callaghan, J. F., & Mark, D. M. (1984). The extraction of drainage networks from digital elevation data. *Computer Vision, Graphics, and Image Processing, 28*(3), 323-344. https://doi.org/10.1016/S0734-189X(84)80011-0

Palowitch, J., Tsitsulin, A., Mayer, B., & Perozzi, B. (2022). GraphWorld: Fake Graphs Bring Real Insights for GNNs. *Proceedings of the 28th ACM SIGKDD Conference on Knowledge Discovery and Data Mining*, 3691-3701. https://doi.org/10.1145/3534678.3539203

Raifer, M., Troilo, R., Kowatsch, F., Auer, M., Loos, L., Marx, S., Przybill, K., Fendrich, S., Mocnik, F.-B., & Zipf, A. (2019). OSHDB: A framework for spatio-temporal analysis of OpenStreetMap history data. *Open Geospatial Data, Software and Standards, 4*, 3. https://doi.org/10.1186/s40965-019-0061-3

Runfola, D., Anderson, A., Baier, H., Crittenden, M., Dowker, E., Fuhrig, S., Goodman, S., Grimsley, G., Layko, R., Melville, G., Mulder, M., Oberman, R., Panganiban, J., Peck, A., Seitz, L., Shea, S., Slevin, H., Youngerman, R., & Hobbs, L. (2020). geoBoundaries: A global database of political administrative boundaries. *PLOS ONE, 15*(4), e0231866. https://doi.org/10.1371/journal.pone.0231866

Saito, T., & Rehmsmeier, M. (2015). The Precision-Recall Plot Is More Informative than the ROC Plot When Evaluating Binary Classifiers on Imbalanced Datasets. *PLOS ONE, 10*(3), e0118432. https://doi.org/10.1371/journal.pone.0118432

Vemula, S., Gatti, F., & Jehel, P. (2026). Graph transformer-based flood susceptibility mapping: Application to the French Riviera and railway infrastructure under climate change. *Journal of Hydrology: Regional Studies, 66*, 103644. https://doi.org/10.1016/j.ejrh.2026.103644

Wang, N., Li, M., Zhang, H., Cheng, W., Du, C., & Lombardo, L. (2025). Susceptibility modeling of hydro-morphological processes considered river topology. *Geo-spatial Information Science, 28*(5), 2652-2671. https://doi.org/10.1080/10095020.2024.2440614

Wang, Z., Lyu, H., & Zhang, C. (2023). Pluvial flood susceptibility mapping for data-scarce urban areas using graph attention network and basic flood conditioning factors. *Geocarto International, 38*(1), 2275692. https://doi.org/10.1080/10106049.2023.2275692

WorldPop. (2025). *Global demographic data: Public release R2024B V1* [Release statement]. WorldPop. https://data.worldpop.org/repo/prj/Global_2015_2030/R2024B/doc/Global2_Release_Statement_R2024B_v1.pdf

Zanaga, D., Van De Kerchove, R., Daems, D., De Keersmaecker, W., Brockmann, C., Kirches, G., Wevers, J., Cartus, O., Santoro, M., Fritz, S., Lesiv, M., Herold, M., Tsendbazar, N.-E., Xu, P., Ramoino, F., & Arino, O. (2022). *ESA WorldCover 10 m 2021 v200* [Data set]. Zenodo. https://doi.org/10.5281/zenodo.7254221